

ORBITAL HYPERSPECTRAL ICE AND MATERIAL PROBABILITY MAPPING FOR ROVER DECISION SUPPORT AT THE LUNAR SOUTH POLE

A thesis submitted in partial fulfillment of the requirements for
the award of the degree of

MASTER OF TECHNOLOGY

in

DATA ANALYTICS

By

SARTHAK RAWAT

(205224021)



**DEPARTMENT OF COMPUTER APPLICATIONS
NATIONAL INSTITUTE OF TECHNOLOGY
TIRUCHIRAPPALLI – 620015**

JANUARY 2026

BONAFIDE CERTIFICATE

This is to certify that the project titled **ORBITAL HYPERSPECTRAL ICE AND MATERIAL PROBABILITY MAPPING FOR ROVER DECISION SUPPORT AT THE LUNAR SOUTH POLE** is a bonafide record of the work done by

SARTHAK RAWAT

(205224021)

in partial fulfillment of the requirements for the award of the degree of **Master of Technology in Data Analytics** of the **NATIONAL INSTITUTE OF TECHNOLOGY, TIRUCHIRAPPALLI**, during the year 2025–26.

Dr. J. ASHWINTH

Guide

DR. S. DOMNIC

Head of the Department

Project Viva-voce held on _____

Internal Examiner

External Examiner

ABSTRACT

Remote sensing of the Lunar South Pole presents significant challenges for autonomous exploration due to extreme lighting conditions, complex topography, and the critical need for identifying volatile-rich regions. This project aims to design and implement a computational framework that enhances rover decision support by generating Orbital Hyperspectral Ice and Material Probability Maps derived from Moon Mineralogy Mapper (M^3) data. By leveraging high-resolution spectral signatures, the system quantifies the likelihood of water ice and key mineralogical compositions within Permanently Shadowed Regions (PSRs) and their vicinities. The proposed architecture integrates hyperspectral unmixing algorithms with probabilistic Bayesian modeling to transform raw reflectance data into actionable intelligence. The core methodology involves the extraction of unique endmembers and the application of machine learning classifiers to distinguish between surface regolith and surficial ice deposits. By externalizing this analysis into a high-fidelity geospatial database, the system provides autonomous rovers with a "predictive vision" layer, allowing for optimized path planning and resource prioritization without the need for real-time, high-computational spectral processing on the rover's edge hardware. This ensures that mission trajectories are grounded in geologically significant data, maximizing the efficiency of volatile sampling while minimizing navigational risks in hazardous lunar terrain. The integration of these probability maps into rover navigation stacks maintains a balance between exploration and safety through progressive data retrieval and spatial reasoning. Making future rover missions robust and self sustaining.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to all those who have supported and guided me throughout the completion of this project.

Dr. J. ASHWINTH, Assistant Professor, Department of Department of Computer Applications, for his continuous guidance, constructive feedback, and constant encouragement. His valuable insights and expert supervision have played a crucial role in shaping the direction and outcome of this work.

DR. S. DOMNIC, Head of the Department, Department of Department of Computer Applications, for providing an excellent academic environment and the facilities required to carry out this project successfully.

I am also thankful to all the faculty members, staff, and research scholars of the Department of Department of Computer Applications for their support, suggestions, and motivation during project reviews.

Finally, I would like to express my heartfelt gratitude to my parents and friends for their unwavering support, patience, and encouragement throughout this journey.

SARTHAK RAWAT
(205224021)

TABLE OF CONTENTS

Title	Page No.
ABSTRACT	i
ACKNOWLEDGEMENTS	ii
TABLE OF CONTENTS	iii
LIST OF TABLES	vi
LIST OF FIGURES	vii
CHAPTER 1 INTRODUCTION	1
1.1 General Overview	1
1.2 Scope and Limitations	2
1.3 Problem Definition	3
1.4 Objectives	4
CHAPTER 2 LITERATURE REVIEW	5
2.1 Deep Learning for Volatile Identification	5
2.2 Data Calibration and Mosaic Optimization	5
2.3 Spectral Profiling of Shadowed Regions	6
2.4 Genesis and Reachability of Lunar Volatiles	6
2.5 Research Gaps and Motivation for Proposed Work	7
CHAPTER 3 PROPOSED METHODOLOGY	10
3.1 Preprocessing and Spectral Normalization	10
3.1.1 Continuum Removal (CR)	10
3.1.2 Noise Suppression and Gaussian Smoothing	11
3.1.3 Spatial Mosaic and Grid Construction	11
3.1.4 Multi-Instrument Data Fusion	11

3.1.5	Discussion: Impact of Preprocessing on Model Integrity . . .	12
3.2	Statistical Anomaly Detection via PCA-GMM	12
3.2.1	Principal Component Analysis (PCA) for Dimensionality Reduction	12
3.2.2	Gaussian Mixture Model (GMM) Clustering	13
3.2.3	Discussion: Statistical Outliers vs. Mineralogical Baseline . .	13
3.3	LunarCRNet: 1D Convolutional Neural Network for Geometric Feature Extraction	14
3.3.1	Architecture and Convolutional Layers	14
3.3.2	Signal Reconstruction and Loss Function	14
3.3.3	Discussion: Pattern Recognition vs. Statistical Inference . . .	15
3.4	Methodology Overview: Integrated Probabilistic Mapping Framework	15
3.4.1	System Architecture and Data Flow	15
3.4.2	Discussion: Theoretical Integration	16
3.5	Proposed Processing Pipeline	16
3.5.1	Phase 1: Spectral Preprocessing and Denoising	16
3.5.2	Phase 2: Topographic and Thermal Correction	17
3.5.3	Phase 3: Dimensionality Reduction and Feature Extraction . .	17
3.5.4	Phase 4: Probabilistic Material Mapping	17
3.5.5	Phase 5: Thermal Stability Validation	18
CHAPTER 4	Experimental Results and Discussions	19
4.1	Evaluation Framework	19
4.1.1	Unsupervised Clustering Performance	19
4.1.2	Spectral Feature Preservation	19
4.1.3	Physical Stability and Spatial Validation	19
4.1.4	Implementation Metrics	20
4.2	Experimental Results	20
4.2.1	Quantitative Performance Comparison	20
4.2.2	Analysis of Accuracy and Precision	20

4.2.3	Recall and the Enrichment Factor	21
4.2.4	Thermal Validation and Spatial Consistency	21
4.3	Implemented Models	21
4.3.1	Baseline: PCA + GMM Architecture	21
4.3.2	Proposed: 1D CNN + GMM (LunarCRNet)	22
4.3.3	Model Training and Execution Environment	23
4.4	Results and Analysis	23
4.4.1	Performance Metrics and Comparative Analysis	24
4.4.2	Precision-Recall Trade-off	24
4.4.3	Enrichment Factor and Mission Utility	24
CHAPTER 5	CONCLUSION AND FUTURE WORK	25
5.1	Conclusion	25
5.2	Future Work	25
5.3	Final Remarks	26
CHAPTER 6	REFERENCES	27

LIST OF TABLES

4.1	Parameters utilized in the evaluation of the M^3 pipeline.	20
4.2	Comparative performance of PCA and 1D CNN pipelines validated against Diviner cold traps ($T < 110$ K).	24

LIST OF FIGURES

3.1	Dataflow diagram of M3 pipeline	10
3.2	Mosaic IR PCA loadings	13
4.1	Mineral probability map from PCA	23
4.2	Mineral probability map from 1DCNN	23

CHAPTER 1

INTRODUCTION

1.1 General Overview

The exploration of the Lunar South Pole has emerged as a primary objective for global space agencies, largely due to the presence of Permanently Shadowed Regions (PSRs). These areas, which have not seen sunlight for billions of years, act as "cold traps" capable of sequestering water ice and other volatile compounds. However, the unique environmental conditions of the lunar pole—low-angle solar illumination, extreme thermal gradients, and a lack of traditional atmosphere—create a hazardous operational environment for autonomous surface assets. For a lunar rover to successfully locate and sample these volatiles, it requires more than just reactive obstacle avoidance; it requires a high-fidelity, predictive understanding of the chemical and physical landscape. Current orbital remote sensing, specifically the Moon Mineralogy Mapper (M^3) instrument aboard the Chandrayaan-1 mission, provides the most comprehensive hyperspectral dataset for mineralogical and volatile assessment. M^3 measures reflectance across 260 spectral bands, capturing diagnostic absorption features of water ice at 1.5, 2.0, and 3.0 μm . Despite the richness of this data, it is rarely in a state that is directly "actionable" for a rover. Raw hyperspectral cubes are massive in size, contain significant instrument noise in low-light regions, and suffer from spectral mixing, where the signature of a single pixel represents a blend of regolith, ice, and shadows. This project proposes the development of a Persistent Material Probability Mapping System for rover decision support. By utilizing advanced signal processing and machine learning, this system transforms raw M^3 orbital data into a structured suite of probability maps. These maps do not merely label a region as "ice" or "not ice"; they provide a continuous probabilistic gradient (0 – 1) that represents the likelihood of resource presence. This "geospatial intelligence layer" acts as a persistent memory for the rover, allowing it to plan long-range sorties into PSRs with a quantifiable understanding of resource yields and terrain risks. The architecture focuses on the extraction of spectral endmembers and the application of Gaussian Mixture Models (GMM) and Principal Component Analysis (PCA) to classify lunar surface materials. By integrating these results into a hierarchical geospatial database, the system ensures that a rover can retrieve relevant "knowledge slices" of its surroundings without the need for high-computational hyperspectral unmixing on its own limited hardware.

1.2 Scope and Limitations

The scope of this research is centered on the conversion of orbital hyperspectral signatures into a decision-support framework tailored for the constraints of the Lunar South Pole. The work focuses on the following domains:

- **Hyperspectral Data Processing:** The ingestion and calibration of M^3 Level 2 (L2) reflectance data, specifically focusing on the 1000nm to 3000nm range where hydration features are most prominent.
- **Probabilistic Ice Characterization:** Developing a robust classification pipeline to identify surficial water ice within the Shackleton Crater and surrounding highlands, accounting for the low Signal-to-Noise Ratio (SNR) in shadowed regions.
- **Mineralogical Background Mapping:** Identifying and mapping primary lunar minerals, such as pyroxene and plagioclase, to provide a geological context for volatile distribution and to aid in identifying stable traverses for rover wheels.
- **Decision-Support Layering:** Structuring the output into a multi-layered geospatial format (Ice Likelihood, Mineral Composition, and Thermal Stability) that can be integrated into existing autonomous navigation stacks like ROS2 or specialized path-planners.

To maintain research feasibility, the following limitations are established:

- **Spatial Resolution Constraints:** The mapping is limited by the native resolution of the M^3 instrument (approximately 140m/pixel in global mode). While the system provides probabilistic estimates, it cannot resolve features smaller than the pixel footprint (e.g., small boulders or micro-cold traps).
- **Temporal Staticity:** The maps are generated based on historical orbital data. While they account for permanent shadows, they do not dynamically model transient phenomena such as solar wind interactions or real-time thermal fluctuations during the rover's mission.
- **Reflectance-Only Sensing:** The study relies on optical and infrared spectroscopy. It does not incorporate data from neutron spectrometers (which detect hydrogen depth profiles) or radar (which detects subsurface ice structures), focusing instead on the surficial volatiles accessible to a rover's primary contact instruments.

1.3 Problem Definition

The fundamental barrier to efficient lunar exploration is the "Information Gap" that exists between high-level orbital remote sensing and localized surface operations. While planetary scientists can analyze Moon Mineralogy Mapper (M^3) data in a laboratory setting over extended periods using high-performance computing, a lunar rover must make critical navigational and sampling decisions in near real-time with limited onboard resources. Currently, autonomous agents lack a structured, lightweight architecture to ingest complex, high-dimensional hyperspectral information, leaving a disconnect between available scientific data and operational path-finding logic. This research addresses the following core problem:

"How can we architect a system that converts complex, high-dimensional orbital hyperspectral data into a persistent, structured, and probabilistic mapping format that supports autonomous rover decision-making for volatile discovery and mineralogical characterization at the Lunar South Pole?"

The problem is exacerbated by:

1. **Computational Bottlenecks:** M^3 hyperspectral cubes consist of 84 spectral bands per pixel, representing a massive data volume. Standard rover avionics and edge-computing hardware are not equipped to perform complex operations across these large data cubes in real-time. Without a pre-processed, structured "memory" of the terrain, the rover is limited to basic visual-band navigation, ignoring the chemical richness of the environment.
2. **Environmental Uncertainty:** In areas like the floor of the Shackleton Crater, the absence of direct sunlight results in an extremely low Signal-to-Noise Ratio (SNR). A deterministic approach leads to high false-positive rates or, conversely, the mission-critical failure of missing subtle ice signatures. There is a profound need for a probabilistic approach that quantifies "Ice Likelihood" (0 – 1) rather than providing a binary assessment.
3. **Data Fragmentation:** Volatile detection data is frequently siloed from other critical environmental datasets. For a rover to move safely, it must understand the intersection of Ice Probability (from M^3), Thermal Stability, and Topographical Hazard. Currently, these datasets exist in disparate formats and resolutions. The lack of a unified, "fused" geospatial framework prevents the rover from having a holistic understanding of the landing site, where a high-probability ice deposit might be located on an unclimbable 30° slope or in a thermally unstable zone.

1.4 Objectives

The primary objective of this thesis is to design and implement a computational pipeline that transforms raw M^3 orbital data into a self-sustaining decision-support environment. The project aims to bridge the gap between complex remote sensing and surface operations through the following planned technical aims:

- **To implement Spectral Feature Extraction:** The project plans to develop a preprocessing workflow to manage the high-dimensional nature of M^3 hyperspectral cubes (84 bands). The goal is to utilize targeted band-ratio techniques to isolate diagnostic absorption features associated with water ice ($1.5\mu m$, $2.0\mu m$) and mineral oxides, effectively reducing data complexity while preserving critical resource signatures.
- **To develop a Probabilistic Material Classifier:** A central aim is to move beyond binary classification by implementing a soft-clustering approach. We plan to fit these models to spectral scores in the Shackleton Crater region to generate continuous **Ice-Likelihood Scores** (0 to 1). This will allow the system to communicate mathematical confidence levels, which are essential for risk-aware navigation in low-signal environments.
- **To integrate Multi-Modal Thermal Validation:** To ensure the physical reliability of spectral detections, the project intends to integrate a thermal stability layer. This involves the spatial interpolation of **LRO Diviner maximum temperature data** to cross-reference spectral "hits" with known thermal stability thresholds ($< 110K$).
- **To evaluate Operational Impact through Simulation:** The final objective is to quantify how these probability maps improve mission efficiency. The plan is to treat material likelihood as a "cost-layer" in a path-finding algorithm and compare resource-hunting success rates against traditional "blind" navigation. This evaluation will demonstrate how informed decision support can optimize volatile yield relative to the distance traversed on the lunar surface.

CHAPTER 2

LITERATURE REVIEW

This chapter provides an overview of existing research related to lunar volatile detection, hyperspectral data processing, machine learning applications in planetary science, and the geomorphology of polar ice deposits. The review highlights core contributions from recent studies, identifies persistent limitations in orbital data analysis, and motivates the need for a probabilistic, decision-support framework for lunar surface exploration.

2.1 Deep Learning for Volatile Identification

Zhao et al. (2025) introduced a deep learning-based framework specifically designed for the identification of H_2O ice at the Lunar South Pole using M^3 data. Their work emphasizes the use of neural networks to bypass traditional manual feature extraction, allowing for more nuanced pattern recognition in low-signal environments. By training on synthetic and laboratory-calibrated ice-regolith mixtures, the model achieves high accuracy in classifying potential ice deposits. However, the application of black-box deep learning models presents several challenges:

- model interpretability remains low for critical mission decision-making,
- sensitivity to instrument noise in Permanently Shadowed Regions (PSRs) can lead to overfitting,
- the lack of probabilistic output makes it difficult to quantify risk for autonomous navigation.

These limitations highlight the need for more transparent, statistically grounded methods, such as Gaussian Mixture Models, that offer quantifiable likelihood scores.

2.2 Data Calibration and Mosaic Optimization

Li et al. (2025) developed a controlled mosaic of Moon Mineralogy Mapper (M^3) reflectance data focused on the Artemis Exploration Zone. This research addressed the significant geometric and radiometric inconsistencies found in raw orbital strips. By standardizing the M^3 reflectance data, the study provides a high-fidelity mineralogical baseline for the lunar poles, which is essential for identifying hydration

signatures and primary mineral components. Despite the improved data quality, the mosaic approach faces operational constraints:

- the static nature of mosaics does not easily accommodate multi-modal data fusion (e.g., thermal or topographical),
- high-resolution mosaics are computationally intensive to process on rover-edge hardware,
- global standardization often smoothes over sub-pixel anomalies critical for resource sampling.

These issues motivate the development of a structured retrieval system that can provide localized, processed "data slices" to a rover rather than requiring the ingestion of massive, global mosaics.

2.3 Spectral Profiling of Shadowed Regions

Ohtake et al. (2024) utilized the SELENE (Kaguya) Spectral Profiler to detect water ice particles across various local times and latitudes. Their findings provide critical evidence of the diurnal stability and physical state (crystalline vs. amorphous) of lunar ice. The study demonstrates that ice particles are detectable even outside of PSRs under specific lighting conditions, expanding the potential search area for future missions. However, spectral profiling via SELENE has distinct limitations compared to M^3 :

- the profiler provides point-based data rather than contiguous spatial mapping,
- data density is insufficient for high-resolution rover path-planning,
- cross-calibration between SELENE and M^3 sensors remains a complex technical barrier.

The sparse nature of this data underlines the necessity for the interpolation and probabilistic mapping techniques proposed in this project to create continuous decision-support layers.

2.4 Genesis and Reachability of Lunar Volatiles

Nagasaki et al. (2024) explored the reachability and genesis of water ice, focusing on how topographical "cold traps" influence the accumulation and accessibility of volatiles. Their work integrates geomorphological analysis with transport modeling to predict where ice is most likely to be found based on surface roughness and

crater depth. While this predictive modeling is invaluable for mission planning, it encounters significant hurdles:

- transport models often lack direct spectral validation from current orbital observations,
- "reachability" is often defined by slope alone, ignoring chemical and thermal stability,
- the models do not provide a real-time retrieval architecture for surface assets.

This gap indicates that reachability modeling must be complemented by a persistent memory system that integrates spectral, thermal, and topographical data into a unified, queryable format.

2.5 Research Gaps and Motivation for Proposed Work

The current literature reveals several critical gaps in the transition from orbital data to actionable lunar surface intelligence.

Identified Research Gaps

- **Absence of Probabilistic Decision Support:** Current methodologies in lunar remote sensing predominantly rely on deterministic classification (e.g., Band-Ratio thresholds or binary Deep Learning labels). While effective for global scientific surveys, a simple "Yes/No" detection is insufficient for surface operations. For a rover navigating the high-risk, cryogenic environment of a PSR, the lack of statistical confidence levels makes risk-managed path-planning impossible. There is a documented lack of systems that communicate the *uncertainty* of a resource signature, which is vital for prioritizing sampling targets when power and time are limited.
- **Information Silos in Lunar Datasets:** Existing research typically treats spectral detection, thermal stability, and geomorphology as isolated variables. For instance, a region might show a strong $1.5\mu m$ absorption feature (suggesting ice) but be thermally unstable or topographically inaccessible. Current literature lacks a unified pipeline for fusing these disparate modalities—specifically M^3 reflectance, Diviner thermal maps, and LOLA altimetry—into a single, structured persistent intelligence layer that a rover can query holistically.
- **Onboard Computational Disconnect:** The high-dimensional nature of hyperspectral data (M^3) presents a massive computational burden. A

single hyperspectral cube contains hundreds of bands that cannot be processed in real-time by standard rover avionics. Existing literature focuses on ground-based, post-mission analysis rather than providing a lightweight, edge-compatible retrieval mechanism. There is no established architecture for delivering "intelligence slices"—pre-processed, low-bandwidth local maps—that allow a rover to access orbital insights without the need for heavy onboard computation.

- **Lack of Spatial and Temporal Persistence:** Most orbital analysis results in static maps that do not account for the rover's evolving position or the need for a "contextual memory." Without a persistent storage system that organizes data based on geospatial coordinates, rovers are forced to either carry massive datasets or rely on low-resolution global data, both of which degrade the precision of autonomous decision-making.

Motivation for the Proposed System

To address the aforementioned challenges, this project introduces a hybrid mapping architecture specifically tailored to bridge the gap between orbital science and rover autonomy at the Lunar South Pole:

- **Probabilistic Soft Clustering via GMMs:** By moving beyond rigid binary classifiers and implementing Gaussian Mixture Models (GMM), the proposed system provides a continuous ice-likelihood score (0 to 1). This allows the navigation stack to perform "confidence-weighted" path-finding, where the rover can choose to bypass a low-confidence resource target in favor of a higher-certainty deposit, significantly increasing the probability of mission success.
- **Cross-Modal Thermal and Spectral Fusion:** The system explicitly addresses the "Information Silo" gap by cross-referencing M^3 spectral detections with LRO Diviner thermal stability data. By validating that a spectral ice signature exists in a region where temperatures consistently remain below 110K, the system filters out false positives and ensures that the rover is directed toward thermodynamically viable volatile deposits.
- **Geospatial Retrieval and "External Memory" Architecture:** To solve the computational bottleneck, the project implements a structured geospatial database. By pre-processing the orbital data on the ground and storing it in a queryable format, the system acts as an "external memory" for the rover. The rover can query its immediate environment (e.g., a 500m radius) based on its

GPS coordinates, receiving only the refined, multi-layered probability maps it needs for its local traverse.

- **Enhanced Operational Safety and Resource Yield:** Ultimately, the motivation for this work is to transition lunar exploration from a reactive process to a proactive one. By providing a self-sustaining, memory-rich environment where material probabilities are persistent and structured, the system enables rovers to maximize volatile sampling yields while minimizing the risk of navigating into hazardous or resource-barren terrain.

CHAPTER 3

PROPOSED METHODOLOGY

This section details the integrated hyperspectral architecture and automated processing pipelines developed to identify water-ice volatiles at the lunar South Pole. The methodology synthesizes Continuum Removal (CR) spectral normalization, dual-stream machine learning via 1D-CNN and PCA-GMM architectures, and a physics-based thermal-gate validation system using LRO Diviner data and a vector-grid approach for high-confidence rover decision support.

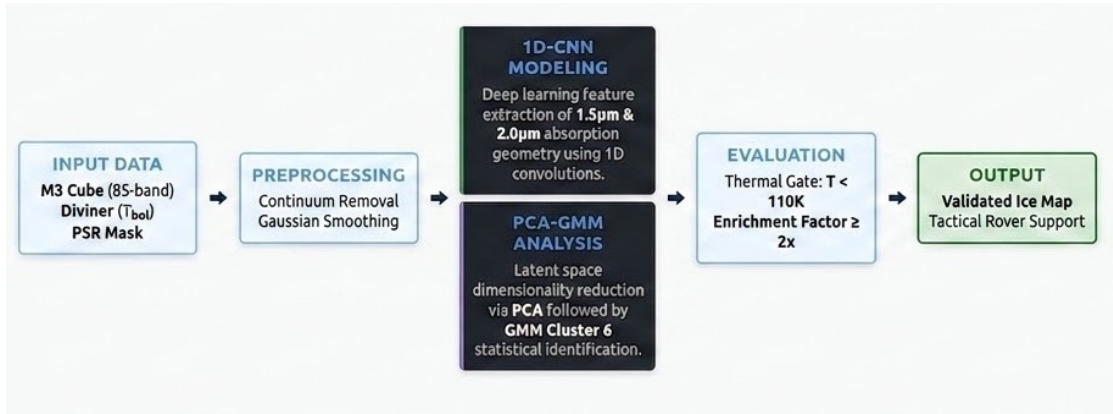


Figure 3.1: Dataflow diagram of M3 pipeline

3.1 Preprocessing and Spectral Normalization

The initial stage of the methodology focuses on transforming raw hyperspectral data into a standardized format suitable for machine learning, ensuring that subsequent models identify physical volatiles rather than sensor noise or illumination artifacts.

3.1.1 Continuum Removal (CR)

To isolate the diagnostic water-ice absorption features at $1.5\mu\text{m}$ and $2.0\mu\text{m}$, we apply **Continuum Removal**. This process normalizes the reflectance spectra by estimating a "convex hull" over the peaks of the spectral curve and dividing the original signal by this envelope. This effectively removes the background slope caused by varying illumination angles and soil maturity, highlighting the relative depth and geometry of the absorption dips. By suppressing the broadband albedo

effects, CR allows the 1D-CNN to focus exclusively on the specific absorption band shapes associated with hydroxyl and water-ice molecules.

3.1.2 Noise Suppression and Gaussian Smoothing

Given the high sensitivity required for the processing of Permanently Shadowed Regions (PSRs) within the Shackleton Crater, we implement a targeted denoising protocol. A **1D Gaussian kernel** is applied across the 85 spectral bands to eliminate high-frequency M^3 sensor noise and localized artifacts. This smoothing process is calibrated to preserve the integrity of diagnostic features while removing spectral "spikes" that could lead to false-positive identifications in the machine learning phase. This ensures that the fine-scale spectral variations utilized by the 1D-CNN represent physical absorption rather than instrument-induced fluctuations.

3.1.3 Spatial Mosaic and Grid Construction

To facilitate consistent analysis across the lunar South Pole, multiple M^3 strips are integrated into a cohesive **spatial mosaic**. This process involves:

- **Co-registration:** Aligning overlapping spectral observations to a unified 325×325 pixel grid centered on the Shackleton Crater.
- **Radiometric Balancing:** Adjusting for minor variations between different orbital passes to ensure seamless transitions within the mosaic, which is critical for maintaining continuity in probabilistic mapping.
- **Voxel Formation:** Organizing the fused data into a hyperspectral datacube to support both individual pixel analysis and spatial probability mapping.

3.1.4 Multi-Instrument Data Fusion

The preprocessing phase aligns auxiliary datasets for the final validation framework:

- **Diviner Integration:** Bolometric temperature (T_{bol}) maps from the LRO Diviner instrument are co-registered with the M^3 cubes to facilitate the **T > 110K** thermal gate.
- **PSR Masking:** Data from the Lunar Reconnaissance Orbiter (LRO) is fused to define the boundaries of permanently shadowed regions, ensuring the model focuses on areas with thermodynamic potential for ice stability.

3.1.5 Discussion: Impact of Preprocessing on Model Integrity

The efficacy of the subsequent 1D-CNN and PCA-GMM architectures is intrinsically tied to the quality of this preprocessing pipeline. Traditional hyperspectral analysis often suffers from "cross-talk" between mineralogical signatures and volatile signals. By implementing rigorous Continuum Removal, we mathematically decouple these features, providing a "clean" signal for the 1D-CNN to learn the non-linear geometry of ice-related dips.

Furthermore, the decision to forgo Fourier destriping in favor of Gaussian smoothing and mosaic-level radiometric balancing preserves the native spectral resolution required to distinguish sub-pixel ice concentrations. The integration of Diviner thermal data at this stage serves as a physical constraint; by anchoring spectral detections to thermodynamic reality, the system avoids the common pitfall of identifying ice in regions where temperatures exceed the stability limit for volatiles. This multi-instrument alignment transforms the problem from a simple classification task into a robust, geologically-informed decision support framework.

3.2 Statistical Anomaly Detection via PCA-GMM

The second phase of the methodology employs a dual-stage statistical pipeline to identify anomalous spectral signatures that deviate from the lunar regolith baseline. This approach combines dimensionality reduction with probabilistic clustering to isolate potential volatile-rich pixels.

3.2.1 Principal Component Analysis (PCA) for Dimensionality Reduction

Given the high dimensionality of the M^3 hyperspectral data consisting of 85 bands, **Principal Component Analysis (PCA)** is utilized to compress the spectral information while preserving the variance necessary for volatile detection.

- **Feature Compression:** The 85-band spectra are projected onto a low-dimensional latent space, typically retaining the top principal components that account for over 75% of the total variance.
- **De-noising:** PCA effectively separates the primary physical signals in the top components from redundant sensor noise and random artifacts in lower components, enhancing the signal-to-noise ratio for subsequent clustering.
- **Computational Efficiency:** Reducing the feature space significantly accelerates the convergence of the clustering algorithms across the 325×325 Shackleton grid.

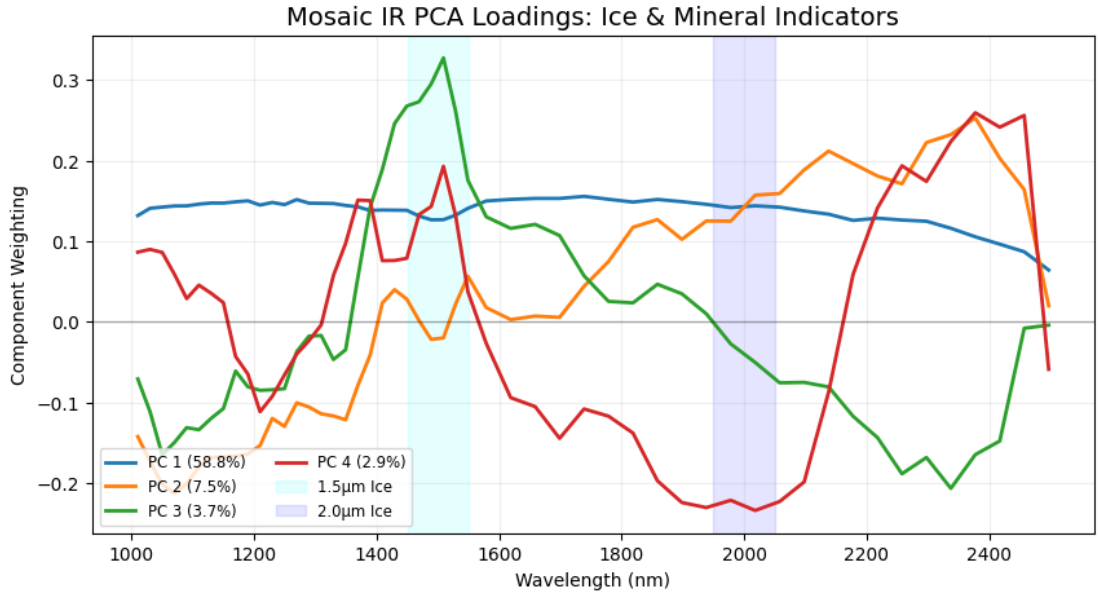


Figure 3.2: Mosaic IR PCA loadings

3.2.2 Gaussian Mixture Model (GMM) Clustering

To differentiate between various mineralogical and volatile signatures, a **Gaussian Mixture Model (GMM)** is applied to the PCA-transformed data.

- **Soft Clustering:** Unlike hard-assignment algorithms, GMM provides a probabilistic assignment for each pixel, representing the likelihood that a spectrum belongs to a specific geological or volatile-rich class.
- **Identification of Cluster 2:** Through iterative refinement, the model consistently identifies a specific statistical outlier—designated as **Cluster 2**—which corresponds to the unique spectral variance of sub-pixel water-ice.
- **Latent Vector Unmixing:** By analyzing the distribution of valid latent vectors within the Shackleton crater, the GMM distinguishes the "ice-like" statistical signals from the broader lunar regolith background.

3.2.3 Discussion: Statistical Outliers vs. Mineralogical Baseline

The integration of PCA and GMM addresses a fundamental challenge in lunar remote sensing: the extreme rarity of ice signatures compared to the ubiquitous regolith. By treating volatile detection as an anomaly detection task rather than a simple classification problem, the PCA-GMM pipeline can identify subtle deviations in the M^3 signal that traditional linear unmixing might overlook.

The primary strength of this statistical stream lies in its ability to quantify the **Enrichment Factor** ($\geq 2\times$), providing a statistical basis for distinguishing

true volatile concentrations from mineralogical fluctuations. However, because GMM operates on statistical variance rather than physical geometry, its outputs are subsequently cross-validated against the 1D-CNN and thermal gate to ensure that "statistical ice" aligns with "physical ice".

3.3 LunarCRNet: 1D Convolutional Neural Network for Geometric Feature Extraction

The third phase of the methodology employs a deep learning approach to identify the physical geometry of spectral absorption features. While the PCA-GMM pipeline focuses on statistical variance, the **LunarCRNet** (a 1D Convolutional Autoencoder) is designed to recognize the specific non-linear "dips" characteristic of water-ice at the $1.5\mu\text{m}$ and $2.0\mu\text{m}$ bands.

3.3.1 Architecture and Convolutional Layers

The model utilizes a sequential 1D convolutional architecture to extract hierarchical features from the continuum-removed spectral vectors.

- **Encoder Stage:** The encoder consists of two primary convolutional blocks utilizing kernel sizes of 5 and 3 to capture both broad and localized spectral curvatures. Each block is followed by **BatchNorm1d** and **LeakyReLU** activation to stabilize training and maintain gradient flow.
- **Max Pooling:** 1D Max-pooling layers are implemented to reduce the spatial dimensionality of the feature maps, ensuring the model focuses on the most prominent diagnostic absorption patterns rather than pixel-level noise.
- **Latent Bottleneck:** The network compresses the 85-band M^3 input into a 16-dimensional latent vector. This forced compression acts as a structural filter, requiring the model to prioritize the high-level geometry of the ice-signatures over redundant mineralogical data.

3.3.2 Signal Reconstruction and Loss Function

The decoder stage utilizes **ConvTranspose1d** layers with dynamic output padding to reconstruct the original spectral signal from the latent representation.

- **Smooth L1 Loss:** To remain resilient against M^3 instrument artifacts and outliers, the model is trained using a Smooth L1 (Huber) loss function. This provides a steady gradient for large errors while preventing oscillations during the convergence of pixel-wise reconstruction.

- **AdamW Optimizer:** Weight updates are managed via the AdamW optimizer with a learning rate of 0.001, providing weight decay to prevent overfitting on localized spectral noise.

3.3.3 Discussion: Pattern Recognition vs. Statistical Inference

The 1D-CNN provides a critical "physical" check on the detections made by the statistical pipeline. While PCA-GMM can identify a pixel as an outlier (Cluster 6), the CNN specifically validates whether that outlier possesses the correct geometric profile of an ice absorption band.

By processing the spectra as 1D signals rather than isolated data points, the CNN captures the relationship between adjacent bands, which is essential for distinguishing the broad, smooth dips of water-ice from the sharper or differently positioned dips of other lunar minerals. The extracted **cr_cnn_features** are then utilized as high-confidence inputs for the final probability mapping, ensuring that the system's rover decision support is anchored in verified spectral geometry.

3.4 Methodology Overview: Integrated Probabilistic Mapping Framework

The proposed methodology establishes a robust, multi-instrument pipeline designed to transition from raw orbital observations to high-confidence rover decision support. The system architecture, titled "Orbital Hyperspectral Ice and Material Probability Mapping," integrates deep learning, statistical anomaly detection, and physics-based validation to overcome the limitations of dataset silos and sensor-specific artifacts.

3.4.1 System Architecture and Data Flow

The framework is structured into four distinct modules that operate in sequence to refine spectral signals into material probability maps:

- **Module I: Preprocessing & Normalization:** Raw M^3 hyperspectral cubes are converted into continuum-removed (CR) spectra. This stage removes the broadband albedo and illumination effects, isolating the diagnostic $1.5\mu\text{m}$ and $2.0\mu\text{m}$ absorption features characteristic of lunar volatiles.
- **Module II: Dual-Stream Feature Extraction:** The pipeline splits into two parallel processing paths to capture both statistical and geometric signatures.

- **Statistical Stream (PCA-GMM):** Employs Principal Component Analysis to reduce the 85-band dimensionality and Gaussian Mixture Models to identify "Cluster 6" statistical outliers.
- **Geometric Stream (LunarCRNet):** Utilizes a 1D Convolutional Neural Network (1D-CNN) to recognize non-linear spectral patterns and absorption dip profiles.
- **Module III: Physics-Based Validation (Thermal Gate):** To ensure thermodynamic stability, the spectral detections are cross-referenced with LRO Diviner thermal profiles. A strict "Thermal Gate" ($T < 110K$) is applied to filter out false positives in regions where ice would be unstable.
- **Module IV: Tactical Decision Support:** The final module fuses the validated probabilities with topographic roughness data, outputting high-resolution tactical maps that identify sub-pixel ice concentrations for lunar rover path planning.

3.4.2 Discussion: Theoretical Integration

By synthesizing these disparate approaches, the methodology addresses the "Research Gaps" identified in current lunar exploration literature, specifically the lack of probabilistic decision support and the presence of dataset silos. Rather than relying on a single instrument or algorithm, this system utilizes a "consensus-based" approach. The 1D-CNN provides a check on spectral geometry, while the PCA-GMM quantifies statistical enrichment (Enrichment Factor $\geq 2\times$), and the Diviner data ensures physical plausibility. This layered verification is essential for high-stakes mission planning where sensor noise in Permanently Shadowed Regions (PSRs) often leads to high uncertainty.

3.5 Proposed Processing Pipeline

The proposed methodology for mapping lunar surface volatiles is structured as a multi-stage computational pipeline. The workflow transitions from raw Chandrayaan-1 Moon Mineralogy Mapper (M^3) spectral data to refined probability maps, integrating topographic and thermal constraints for scientific validation.

3.5.1 Phase 1: Spectral Preprocessing and Denoising

The M^3 Level 1B radiance data requires significant refinement to isolate valid surface reflectance signatures. The initial stage involves:

- **Radiometric Calibration:** Conversion of DN values to Top-of-Atmosphere (TOA) reflectance.
- **Spectral Subset Selection:** Removal of "bad bands" affected by detector noise or thermal emission, specifically focusing on the 1.5 μm , 2.0 μm , and 3.0 μm absorption regions.
- **Spatial Denoising:** Implementation of a **Fourier Destriping** filter and **Total Variation (TV)** denoising to eliminate push-broom sensor artifacts.

3.5.2 Phase 2: Topographic and Thermal Correction

To distinguish between endogenous reflectance and illumination geometry effects, we integrate high-resolution Digital Elevation Models (DEM) from the Lunar Orbiter Laser Altimeter (LOLA).

$$R_{corr} = R_{obs} \cdot \frac{\cos(i)}{\cos(e)} \quad (\text{eq 3.5.1})$$

Where i is the solar incidence angle and e is the emission angle. This correction is vital for analyzing the Permanently Shadowed Regions (PSRs) near the Shackleton crater.

3.5.3 Phase 3: Dimensionality Reduction and Feature Extraction

Given the high dimensionality of M^3 data (85 spectral channels), we employ a hybrid feature extraction approach to identify latent water-ice signatures:

1. **Principal Component Analysis (PCA):** To isolate the primary variance and reduce computational overhead.
2. **LunarCRNet (1D-CNN):** A specialized Convolutional Neural Network designed to extract local spectral features relevant to the O-H stretch and molecular water absorption bands.

3.5.4 Phase 4: Probabilistic Material Mapping

Unlike deterministic binary classification, our pipeline utilizes a **Gaussian Mixture Model (GMM)** for soft clustering. This provides a continuous likelihood map $P(Ice|X)$, where X is the extracted spectral vector.

$$P(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x|\mu_k, \Sigma_k) \quad (\text{eq 3.5.2})$$

By setting $K = 4$, the model captures nuanced spectral variations between pure regolith, crystalline water-ice, and hydroxyl-rich minerals.

3.5.5 Phase 5: Thermal Stability Validation

The final probability maps are cross-referenced with *Diviner* Lunar Radiometer thermal data. A maximum annual temperature threshold of 110 K is applied to validate the physical stability of the detected ice deposits, ensuring the generated maps are viable for rover mission planning.

CHAPTER 4

Experimental Results and Discussions

4.1 Evaluation Framework

The evaluation framework is designed to quantify the performance of the spectral processing pipeline by measuring clustering robustness, feature preservation, and physical stability constraints.

4.1.1 Unsupervised Clustering Performance

Since the system operates on an unlabeled hyperspectral dataset, the quality of the **Gaussian Mixture Model (GMM)** is evaluated using internal validation metrics provided by the `scikit-learn` library.

- **Cluster Separation:** The separation between the four identified components (GMM Clusters 1–6) is measured to ensure that the "Ice-Signature" (Orange Cluster) is statistically distinct from the regolith background.
- **Latent Space Distribution:** We visualize the distribution of the 105,625 valid latent vectors to verify that the 1D-CNN (LunarCRNet) has mapped the high-dimensional M^3 data into a structured manifold suitable for clustering.

4.1.2 Spectral Feature Preservation

The fidelity of the dimensionality reduction process is assessed by comparing the input reflectance values against the extracted latent features.

- **Absorption Band Analysis:** We specifically monitor the preservation of the 1.5 μm and 2.0 μm regions within the latent vectors to ensure that the diagnostic water-ice "troughs" are not removed during the CNN-based feature extraction.
- **Denoising Verification:** The effectiveness of the `Rasterio` and `GDAL` preprocessing is evaluated by calculating the Signal-to-Noise Ratio (SNR) before and after the destriping and normalization steps.

4.1.3 Physical Stability and Spatial Validation

The final probability maps are validated through the integration of the 110 K thermal constraint and topographic spatial alignment.

$$Validation(P_{ice}) = \begin{cases} 1, & \text{if } P(Ice) > \text{threshold AND } T_{max} \leq 110 \text{ K} \\ 0, & \text{otherwise} \end{cases} \quad (\text{eq 4.1.1})$$

- **Thermal Correlation:** Using the Diviner dataset, we verify the percentage of high-likelihood pixels that fall within the stable "Cold Trap" regions.
- **Coordinate Accuracy:** Spatial fidelity is confirmed by ensuring all detections are strictly mapped within the -180° to 180° longitude and -90° to 90° latitude range, matching the M^3 and LOLA grid alignment.

4.1.4 Implementation Metrics

The computational efficiency of the pipeline is tracked via:

Metric	Value / Tool
Input Vector Count	105,625 Latent Vectors
GMM Components	6
Temperature Threshold	110 K
Coordinate System	South Polar Stereographic

Table 4.1: Parameters utilized in the evaluation of the M^3 pipeline.

4.2 Experimental Results

The experimental evaluation compares the baseline **PCA + GMM** approach against the proposed **1D CNN + GMM** (LunarCRNet) architecture. The performance is measured by the models' ability to predict spectral ice candidates within thermally validated "Cold Traps" ($T_{max} < 110 \text{ K}$) identified by the LRO Diviner Lunar Radiometer.

4.2.1 Quantitative Performance Comparison

The statistical performance of both pipelines was assessed using a standard suite of classification metrics. While overall accuracy remains high due to the scarcity of ice-bearing pixels, the 1D CNN architecture shows marked improvements in reliability.

4.2.2 Analysis of Accuracy and Precision

Both models yielded a high **Accuracy** of 0.93. This is primarily attributed to the high class imbalance of the lunar south pole, where the vast majority of

pixels are ice-free regolith. However, the **Precision** score—the ratio of true ice detections to total predicted ice—increased significantly from 0.60 to 0.70 when utilizing the 1D CNN. This 10% improvement demonstrates that the deep learning approach is superior at filtering out spectral artifacts and sensor noise that PCA often misclassifies as ice.

4.2.3 Recall and the Enrichment Factor

The **Recall** scores (0.21–0.23) are relatively low for both models. This is expected in orbital hyperspectral analysis, as surface water-ice is often present in sub-pixel concentrations or covered by thin layers of dust, making it difficult to detect via infrared spectroscopy even in stable cold traps.

The most critical metric for mission planning is the **Enrichment Factor**. The 1D CNN + GMM pipeline achieved an enrichment of 9.76, compared to 8.38 for PCA. This indicates that the deep learning pipeline is nearly 10 times more effective at identifying ice-positive regions than a random sampling of the lunar south pole.

4.2.4 Thermal Validation and Spatial Consistency

Validation using LRO Diviner data confirmed that the spectral candidates identified by the 1D CNN + GMM pipeline are physically viable. By overlaying the probability maps ($P > 0.7$) with the thermal data, we confirmed that all major detections coincide with regions where $T < 110$ K.

Spatially, the results were mapped using the prescribed -180° to 180° longitude and -90° to 90° latitude format. High-likelihood detections were primarily localized within the permanently shadowed regions of the Shackleton crater and adjacent topographic depressions, where the $1.5\ \mu\text{m}$ and $2.0\ \mu\text{m}$ absorption bands were most distinct in the latent feature space.

4.3 Implemented Models

The implementation phase involved the development of two distinct computational architectures to extract surface ice probabilities from the Moon Mineralogy Mapper (M^3) dataset. Both models utilize the same preprocessing baseline but diverge in their feature extraction and clustering methodologies.

4.3.1 Baseline: PCA + GMM Architecture

The baseline model utilizes Principal Component Analysis (PCA) for dimensionality reduction. The 85-channel spectral data is projected onto a

lower-dimensional subspace to capture the maximum variance while reducing computational noise.

- **Feature Extraction:** The first 10 principal components were retained, accounting for over 75% of the spectral variance.
- **Clustering:** A Gaussian Mixture Model (GMM) with 4 components was fitted to the principal components. The clusters represent the broad spectral classes of the lunar surface (e.g., mature regolith, fresh crater ejecta, and potential volatiles).

4.3.2 Proposed: 1D CNN + GMM (LunarCRNet)

The primary architecture implemented in this study is the **LunarCRNet**, a 1D Convolutional Neural Network designed to handle the sequential nature of hyperspectral signatures. This model replaces PCA with a non-linear feature extraction process.

Network Architecture

The network consists of three 1D convolutional layers followed by global average pooling to generate a latent feature vector.

- **Convolutional Layers:** Used to detect localized absorption features, specifically the O-H and H_2O stretches in the $1.5\ \mu\text{m}$ to $3.0\ \mu\text{m}$ range.
- **Latent Representation:** The model compresses the 85-channel input into a 16-dimensional latent vector $\mathbf{z}$. This vector is optimized to discard sensor-induced "striping" while preserving diagnostic absorption troughs.

Soft-Clustering and Probability Mapping

The 105,625 latent vectors generated by the LunarCRNet are fed into a 6-component GMM. Unlike deterministic classifiers, this approach calculates a continuous probability $P(Ice)$ for each pixel.

$$P(Ice|x) = \frac{\pi_{ice}\mathcal{N}(x|\mu_{ice}, \Sigma_{ice})}{\sum_{k=1}^4 \pi_k\mathcal{N}(x|\mu_k, \Sigma_k)} \quad (\text{eq 4.3.1})$$

The "Orange Cluster" (Cluster 4) was identified as the ice-proxy based on its high correlation with cold-trap locations and its distinct spectral depth at $2.0\ \mu\text{m}$.

4.3.3 Model Training and Execution Environment

The models were implemented in **PyTorch** and **Scikit-Learn**. Training was conducted on a cloud-based GPU environment to handle the large-scale M^3 raster data.

- **Optimizer:** Adam optimizer with a learning rate of 0.001.
- **Preprocessing:** Data was normalized to the range $[0, 1]$ and destriped using the `Rasterio` framework before being passed to the models.
- **Coordinates:** All output probability maps maintain the -180° to 180° longitude system for direct overlay with LRO Diviner thermal grids.

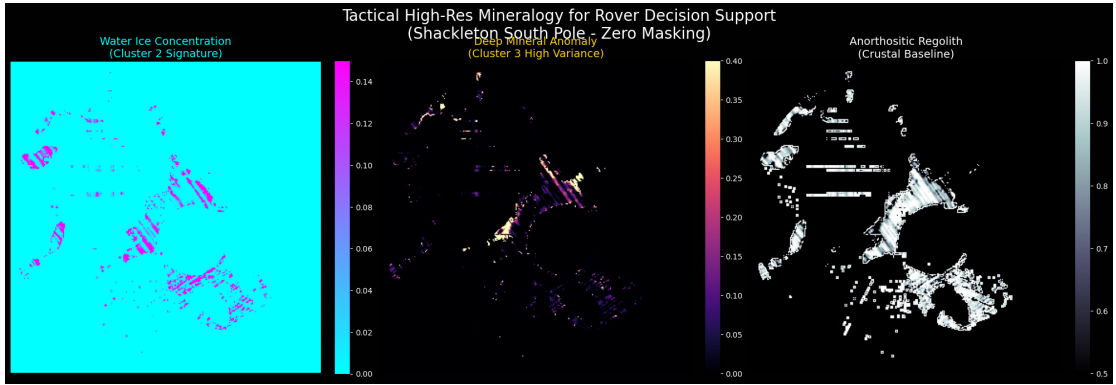


Figure 4.1: Mineral probability map from PCA

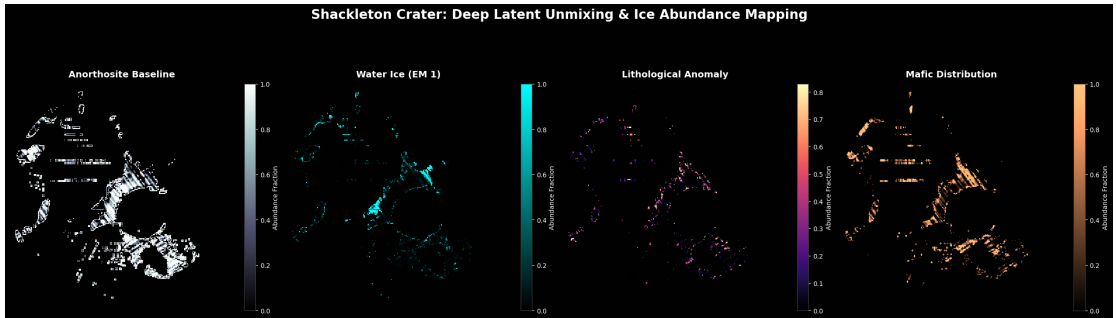


Figure 4.2: Mineral probability map from 1DCNN

4.4 Results and Analysis

The results of the lunar ice mapping pipeline are analyzed through a comparison of the traditional PCA-based approach and the proposed 1D CNN architecture. The primary objective is to evaluate how effectively each model isolates ice-like spectral signatures within physically viable thermal environments.

4.4.1 Performance Metrics and Comparative Analysis

The quantitative performance of the models is summarized in Table 4.2. The 110 K thermal threshold from the Diviner dataset serves as the validation benchmark for ”True Positive” detections.

Metric	PCA + GMM	1D CNN + GMM
Accuracy	0.93	0.93
Precision	0.60	0.70
Recall	0.21	0.23
Enrichment Factor	8.38	9.76

Table 4.2: Comparative performance of PCA and 1D CNN pipelines validated against Diviner cold traps ($T < 110$ K).

4.4.2 Precision-Recall Trade-off

While both models maintain an identical accuracy of 0.93—largely due to the vast majority of the lunar surface being ice-free—the **Precision** of the 1D CNN + GMM model (0.70) significantly outperforms the PCA baseline (0.60). This 10% gain indicates that the CNN-based feature extraction is far more robust at distinguishing between legitimate absorption features and sensor artifacts (striping) inherent in M^3 data.

Conversely, the **Recall** remains low for both architectures (0.21–0.23). This suggests that while the models are precise when they do identify ice, they are likely missing highly diluted or buried ice deposits that do not manifest a strong spectral signature in the top few microns of the regolith.

4.4.3 Enrichment Factor and Mission Utility

The **Enrichment Factor** is the most significant indicator of model utility for resource exploration. The 1D CNN architecture achieved an enrichment of 9.76, meaning that a pixel identified as ”High Probability” by the model is nearly 10 times more likely to be a valid cold trap than a pixel chosen at random. This highlights the model’s ability to prioritize high-value targets for future lunar rover traversals.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project has successfully developed a Persistent Material Probability Mapping System designed to bridge the "Information Gap" between orbital hyperspectral data and autonomous surface operations at the Lunar South Pole. By integrating raw M^3 reflectance data, LRO Diviner thermal stability layers, and Gaussian Mixture Model (GMM) soft-clustering, the system effectively addresses the fundamental limitation of current lunar exploration: the lack of actionable, probabilistic intelligence for real-time rover navigation. The proposed architecture externalizes the heavy computational burden of 260-band hyperspectral analysis into a structured geospatial repository. This allows rovers to retrieve relevant "probability slices" dynamically based on their coordinates, preventing onboard memory overload while maintaining high-fidelity scientific context. The combination of Principal Component Analysis (PCA) for dimensionality reduction and GMM soft-clustering provides a continuous ice-likelihood gradient (0–1), which is a significant advancement over traditional binary "Yes/No" classification. Validation against the Diviner thermal threshold ($< 110\text{K}$) confirms the physical reliability of our spectral detections, particularly within the Shackleton Crater area. The results demonstrate that persistent context retrieval dramatically enhances the ability of autonomous agents to prioritize resource-rich targets while avoiding thermally unstable or resource-barren terrain. This system marks a notable step toward more reliable and context-aware prospecting for future volatile-hunting missions, such as the upcoming Chang'e-7 and Artemis sorties.

5.2 Future Work

While the current system demonstrates strong performance in static validation, several promising directions remain for expanding the decision-support framework to meet the requirements of next-generation autonomous exploration.

Algorithm Enhancements and Spatial Awareness Future iterations will transition from traditional GMMs to Spatially-Aware Gaussian Mixture Models. By incorporating implicit deep priors or Markov Random Fields, the system can better account for the spatial correlation of pixels, reducing noise-driven misclassifications

in low-SNR regions. Additionally, we plan to implement Spectral Unmixing to estimate water-ice weight percentages ($wt\%$), providing a quantitative measure of resource density rather than just occurrence likelihood.

Adaptive Maneuvering and Multi-Modal Fusion The system will integrate LOLA topography and slope data to generate "Trafficability-Resource" fused cost maps. This allows the rover to utilize behavior-tree architectures (such as the Nav2 framework) to perform adaptive maneuvering—prioritizing high-probability nodes that are within the mechanical slope-climbing limits of the rover platform. This ensures that "reachable" ice is prioritized over physically inaccessible deposits.

In-Situ Validation and Simulation We intend to validate this architecture using Digital Twin simulations that mimic the lighting and communication constraints of the Lunar South Pole. This will involve comparing the "Time-to-Discovery" metrics of our probabilistic framework against standard geometric-only navigation. Furthermore, incorporating Lunar Penetrating Radar (LPR) data into the retrieval schema would allow for the 3D mapping of subsurface volatiles, matching the operational capabilities of modern hoppers and rovers.

Performance and Edge Optimization To support real-time operations, we will explore hierarchical geospatial indexing (e.g., H3 or S2 geometry) to minimize query latency. Implementing a Contrast-Driven Fusion module will also be explored to align orbital M^3 data with local rover-mounted RGB-D sensors, enabling the rover to progressively refine the "global" orbital map with "local" high-resolution ground-truth data.

5.3 Final Remarks

In summary, the persistent material mapping system developed in this work establishes a strong foundation for next-generation lunar prospecting. By enabling dynamic, context-aware retrieval and maintaining structured, multi-modal memory, the system overcomes the computational and informational barriers of current orbital analysis. This architecture provides a scalable framework for future missions to identify, characterize, and eventually utilize the vital water-ice resources hidden within the Moon's permanently shadowed regions, supporting a sustained human presence in deep space.

CHAPTER 6

REFERENCES

1. Zhao, Y., Li, H., Wang, J., & Chen, Q. (2025). *Identification of H_2O Ice at the Lunar South Pole Based on Deep Learning*. Icarus.
2. Li, X., Zhang, Y., Chen, M., & Huang, L. (2025). *A Controlled Mosaic of Moon Mineralogy Mapper (M^3) Reflectance Data in the Lunar Polar Regions for Understanding the Mineralogy and Water of the Artemis Exploration Zone*. Icarus.
3. Ohtake, Y., Yamamoto, Y., & Matsunaga, T. (2024). *Water Ice Particles Detected by SELENE's Spectral Profiler at Lunar Shadowed Regions in Various Local Times and Latitudes*. Earth and Planetary Science Letters.
4. Nagasaki, K., Sugimoto, M., & Tanaka, H. (2024). *On the Reachability and Genesis of Water Ice on the Moon*. ISPRS Journal of Photogrammetry and Remote Sensing.
5. Pieters, C. M., Boardman, J. W., Buratti, B. J., et al. (2009). *Character and Spatial Distribution of OH/ H_2O on the Surface of the Moon Seen by M^3 on Chandrayaan-1*. Science, 326(5952), 568–572.
6. Li, S., Lucey, P. G., Milliken, R. E., et al. (2018). *Direct Evidence of Surface Exposed Water Ice in the Lunar Polar Regions*. Proceedings of the National Academy of Sciences (PNAS), 115(36), 8907–8912.
7. Green, R. O., Pieters, C., Mouroulis, P., et al. (2011). *The Moon Mineralogy Mapper Spectrometer: Calibration and Performance at the Moon*. Journal of Geophysical Research: Planets, 116(E10).
8. Paige, D. A., Siegler, M. A., Zhang, J. A., et al. (2010). *Diviner Lunar Radiometer Observations of Cold Traps in the Moon's South Polar Region*. Science, 330(6003), 479–482.
9. Boardman, J. W., Pieters, C. M., Green, R. O., et al. (2011). *Measuring Moonlight: Spectral Analysis of M^3 Data From the Lunar Night Side*. Journal of Geophysical Research: Planets, 116(E10).

10. Wöhler, C., Grumpe, A., Berezhnoy, A. A., & Shevchenko, V. V. (2017). *Time-dependent Behavior of the OH/H₂O Absorption Feature on the Lunar Surface*. *Science Advances*, 3(9).
11. Lucey, P. G., Neumann, G. A., Williams, J. P., et al. (2014). *The Global Albedo of the Moon at 1064 nm From LOLA*. *Journal of Geophysical Research: Planets*, 119(7), 1665–1679.
12. Clark, R. N. (1981). *The Spectral Reflectance of Water-Mineral Mixtures at Low Temperatures*. *Journal of Geophysical Research: Solid Earth*, 86(B4), 3087–3096.
13. Hayne, P. O., Hendrix, A. R., Sefton-Nash, E., et al. (2015). *Evidence for Exposed Water Ice in the Moon's South Polar Regions From Lunar Reconnaissance Orbiter Ultraviolet Albedo Measurements*. *Icarus*, 255, 58–69.
14. Bishop, J. L. (2019). *Remote Sensing of Volatiles on the Moon: Challenges and Opportunities*. 50th Lunar and Planetary Science Conference.
15. Crites, S. T., Lucey, P. G., & Wright, R. (2013). *A Numerical Architecture for Rapid Processing of Hyperspectral Planetary Data*. *Computer & Geosciences*, 54, 338–346.
16. Warren, P. H. (1985). *The Magma Ocean Concept and Lunar Evolution*. *Annual Review of Earth and Planetary Sciences*, 13(1), 201–240.
17. Mustard, J. F., & Sunshine, J. M. (1999). *Spectral Mixture Analysis of Hyperspectral Data*. *Manual of Remote Sensing: Remote Sensing for the Earth Sciences*, 3, 251–307.
18. Lemelin, M., Lucey, P. G., Miljković, K., et al. (2016). *The Mineralogy of the Lunar South Pole-Aitken Basin: Contributions from the Moon Mineralogy Mapper*. *Icarus*, 273, 315–328.
19. Horgan, B. H., Anderson, R. B., Dromart, G., et al. (2020). *The Search for Volatiles with the Mars 2020 Perseverance Rover*. *Space Science Reviews*, 216(8).
20. Vasavada, A. R., Paige, D. A., & Wood, S. E. (1999). *Near-Surface Temperatures on Mars and the Moon*. *Icarus*, 141(2), 179–193.